Gromit Yeuk-Yin CHAN

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RESEARCH INTEREST

Visualization, Data Management, Human Computer Interaction

EDUCATION

2016 - 2021 **New York University**
Ph.D in Computer Science
Advisors: *Cláudio T. Silva, Juliana Freire*
Committee: *Enrico Bertini, Luis Gustavo Nonato, Arvind Satyanarayan*

2012 - 2016 **The Hong Kong University of Science and Technology**
BEng in Computer Engineering and BBA in General Business Management
Advisor: *Huamin Qu*

2005 - 2012 **Queen's College, Hong Kong**
Hong Kong Advanced Level Examination: *Distinction in Physics*

PROFESSIONAL EXPERIENCE

2019 - Now Senior Research Scientist, **Adobe Research**, San Jose, CA

- Previous titles: Research Scientist 2 (2023-2026), Research Scientist (2021-23), Research Intern (2019, 2020)

2019 Feb - Apr Visiting Scholar, **diNo Research Group (Paris Descartes University)**, Paris, France

2017 Jun- Sep Research Intern, **Bosch Research**, Palo Alto, CA

2016 Aug - 2021 July Research Assistant, **NYU VIDA (Visualization, Imaging, and Data Analytics Center)**, Brooklyn, NY

2015 Jun - 2016 Aug Undergraduate Researcher, **HKUST VisLab**, Hong Kong

2015 Jan - May IT Consulting Intern, **IBM Global Business Services**, Hong Kong

2014 Jun - Aug Software Engineering Intern, **Orient Overseas Container Line (OOCL)**, Hong Kong

2013 Sep - Dec Design Consulting Intern, **Harman Kardon**, Hong Kong

2013 Jun - July Logistics Intern, **GE Healthcare**, Beijing, China

PUBLICATIONS (★ indicates premier venues) (Ⓢ indicates students I directly supervised)

Conferences and Journals (15 ★, 7 Ⓢ, 6 first-authored, 2 awarded):

32. Steffen Holter Ⓢ, Eunyee Koh, Doga Dogan, and Gromit Yeuk-Yin Chan.
UXCascade: Scalable Usability Testing with Simulated User Agents.
*ACM Symposium on User Interface Software and Technology (**UIST**)*, 2026 ★

31. Yuzhe You Ⓢ, Gromit Yeuk-Yin Chan, Shunan Guo, Anlan Zhang, Eunyee Koh, Jian Zhao, and Tongyu Zhou.
Surprise2Refine: Axis-Based Exploration-To-Refinement for Agent-Assisted Creative Scaffolding.
*ACM Symposium on User Interface Software and Technology (**UIST**)*, 2026 ★

30. Shishi Xiao Ⓢ, Tongyu Zhou, David Laidlaw, and Gromit Yeuk-Yin Chan.
ChArtist: Generating Pictorial Charts with Unified Spatial and Subject Control.
*Computer Vision and Pattern Recognition Conference (**CVPR**)*, 2026 ★

29. Donghoon Shin Ⓢ, Daniel Lee, Gary Hsieh, and Gromit Yeuk-Yin Chan.
PosterMate: Audience-driven Collaborative Persona Agents for Poster Design.
*ACM Symposium on User Interface Software and Technology (**UIST**)*, 2025 ★

28. Gromit Yeuk-Yin Chan, Luis Gustavo Nonato, Themis Palpanas, Cláudio T Silva, and Juliana Freire.
TiVy: Time Series Visual Summary for Scalable Visualization.
*IEEE Transactions on Visualization and Computer Graphics (**TVCG**)*, 2026 (Proceedings of VIS 2025) ★

27. Chenyi Li, Guande Wu, Gromit Yeuk-Yin Chan, Dishita G Turakhia, Sonia Castelo Quispe, Dong Li, Leslie Welch, Claudio Silva, and Jing Qian.
Satori: Towards Proactive AR Assistant with Belief-Desire-Intention User Modeling.
*ACM Conference on Human Factors in Computing Systems (**CHI**)*, 2025 ★

26. Ishika Joshi Ⓢ, Simra Shahid, Shreeya Venneti, Manushree Vasu, Yantao Zheng, Yunyao Li, Balaji Krishnamurthy, and Gromit Yeuk-Yin Chan.
CoPrompter: User-Centric Evaluation of LLM Instruction Alignment for Improved Prompt Engineering.
*Proceedings of the International Conference on Intelligent User Interfaces (**IUI**)*, 2025

25. Haneen Mohammed, Alexander Yao, Charlie Summers, Hongbin Zhong, Gromit Yeuk-Yin Chan, Subrata Mitra, Lampros Flokas, and Eugene Wu.
FaDE: More Than a Million What-ifs Per Second.
*Proceedings of the VLDB Endowment (**VLDB**)*, 2025 ★

24. Ashley Ge Zhang, Victor Soares Bursztyn, Gromit Chan, Shunan Guo, Eunyee Koh, Steve Oney, and Jane Hoffswell.
ConvoMap: Interactive Visualizations for Exploring Complex Conversations in Multi-Agent Systems.
*IEEE Symposium on Visual Languages and Human-Centric Computing (**VL/HCC**)*, 2025

23. Parikshit Solunke, Vitoria Guardieiro, João Rulff, Peter Xenopoulos, Gromit Yeuk-Yin Chan, Brian Barr, Luis Gustavo Nonato, and Claudio Silva.
MOUNTAINEER: Topology-Driven Visual Analytics for Comparing Local Explanations.
*IEEE Transactions on Visualization and Computer Graphics (**TVCG**)*, 2024 ★

22. Tongyu Zhou Ⓢ, Jeff Huang, and Gromit Yeuk-Yin Chan.
Epigraphics: Message-Driven Infographics Authoring.
*ACM Conference on Human Factors in Computing Systems (**CHI**)*, 2024 ★ (**Best Paper Honorable Mention**)

21. Guande Wu, Shunan Guo, Jane Hoffswell, Gromit Yeuk-Yin Chan, Ryan Rossi, and Eunyee Koh.
Data Story Generation via Adaptive Machine-Guided Elicitation of User Feedback.
*IEEE Transactions on Visualization and Computer Graphics (**TVCG**)*, 2024 (Proceedings of VIS 2023) ★

20. Pat Sukprasert Ⓢ, Gromit Yeuk-Yin Chan, Ryan Rossi, Fan Du, and Eunyee Koh.
Discovery and Matching Numerical Attributes in Data Lakes.
*IEEE International Conference on Big Data (**BigData**)*, 2023

19. Chen Chen, Jane Hoffswell, Shunan Guo, Ryan Rossi, Gromit Yeuk-Yin Chan, Fan Du, Eunyee Koh, and Zhicheng Liu.
WhatsNext: Guidance-enriched Exploratory Data Analysis with Interactive, Low-Code Notebooks.
*IEEE Symposium on Visual Languages and Human-Centric Computing (**VL/HCC**)*, 2023

18. Chieh-Yang Huang, Ting-Yao Hsu, Ryan Rossi, Ani Nenkova, Sungchul Kim, Gromit Yeuk-Yin Chan, Eunyee Koh, C Lee Giles, and Ting-Hao Huang.

Summaries as Captions: Generating Figure Captions for Scientific Documents with Automated Text Summarization.
*International Natural Language Generation Conference (**INLG**)*, 2023(**Best Paper**)

17. Jun Yuan, Gromit Yeuk-Yin Chan, Brian Barr, Kyle Overton, Kim Rees, Luis Gustavo Nonato, Enrico Bertini, and Claudio T Silva. SUBPLEX: A Visual Analytics Approach to Understand Local Model Explanations At the Subpopulation Level.
*IEEE Computer Graphics and Applications (**CG&A**)*, 2022

16. Gromit Yeuk-Yin Chan, Tung Mai, Anup B Rao, Ryan A Rossi, Fan Du, Cláudio T Silva, and Juliana Freire.
Interactive Audience Expansion On Large Scale Online Visitor Data.
*International Conference on Knowledge Discovery and Data Mining (**KDD**)*, 2021 ★

15. Gromit Yeuk-Yin Chan, Fan Du, Ryan A Rossi, Anup B Rao, Eunyee Koh, Cláudio T Silva, and Juliana Freire.
Real-Time Clustering for Large Sparse Online Visitor Data.
*The Web Conference (**WWW**)*, 2020 (Accepted for **Oral Presentation**) ★

14. Gromit Yeuk-Yin Chan, Luis Gustavo Nonato, Alice Chu, Preeti Raghavan, Viswanath Aluru, and Cláudio T Silva.
Motion Browser: Visualizing and Understanding Complex Upper Limb Movement Under Obstetrical Brachial Plexus Injuries.
*IEEE transactions on visualization and computer graphics (**TVCG**)*, 2020 (Proceedings of VIS 2019) ★

13. Gromit Yeuk-Yin Chan, Panpan Xu, Zeng Dai, and Liu Ren.
VIBR: Visualizing Bipartite Relations at Scale with the Minimum Description Length Principle.
*IEEE transactions on visualization and computer graphics (**TVCG**)*, 2019 (Proceedings of VIS 2018) ★

12. Quan Li, Peng Xu, Yeuk-Yin Chan, Yun Wang, Zhipeng Wang, Huamin Qu, and Xiaojuan Ma.
A Visual Analytics Approach for Understanding Reasons behind Snowballing and Comeback in MOBA Games.
*IEEE transactions on visualization and computer graphics (**TVCG**)*, 2017 (Proceedings of VIS 2016) ★

11. Yeuk-Yin Chan and Huamin Qu.
FinaVistory: Using Narrative Visualization to Explain Social and Economic Relationships in Financial News.
*International Conference on Big Data and Smart Computing (**BigComp**)*, 2016

Workshops, Demos and Posters (7 ★, 1 Ⓢ, 2 first-authored, 1 awarded):

10. Xinyu Shi, Shunan Guo, Jane Hoffswell, Gromit Yeuk-Yin Chan, Victor S Bursztyn, Jian Zhao, and Eunyee Koh.
Comprehensive Sketching: Exploring Infographic Design Alternatives in Parallel.
*ACM Conference on Human Factors in Computing Systems (Late Breaking Work) (**CHI LBW**)*, 2025 ★

9. Tongyu Zhou Ⓢ, Gromit Yeuk-Yin Chan, Shunan Guo, Jane Hoffswell, Chang Xiao, Victor Soares Bursztyn, and Eunyee Koh.
Data Pictorial: Deconstructing Raster Images for Data-Aware Animated Vector Posters.
*ACM Symposium on User Interface Software and Technology (**UIST**)*, 2024 ★

8. Alex Tang, Gromit Yeuk-Yin Chan, Ryan Rossi, Chang Xiao, and Eunyee Koh.
Tabular Data to Image Generation: Benchmark Data, Approaches, and Evaluation [Vision Paper].
*IEEE International Conference on Big Data (**BigData**)*, 2023

7. Gromit Yeuk-Yin Chan, Shunan Guo, Caroline Kim, Cole Connelly, Michelle Lee, Andrew Thomson, and Eunyee Koh.
Detecting Email Components with Constraints: Expressive and Extensible Models in Answer Set Programming.
*ACM Conference on Human Factors in Computing Systems (Late Breaking Work) (**CHI LBW**)*, 2023 ★

6. Shubham Agarwal, Gromit Yeuk-Yin Chan, Shaddy Garg, Tong Yu, and Subrata Mitra.
Fast Natural Language Based Data Exploration with Samples.
*ACM International Conference on Management of Data (**SIGMOD**)*, 2023 ★

5. Ryan Aponte, Ryan Rossi, Shunan Guo, Jane Hoffswell, Nedim Lipka, Chang Xiao, Gromit Yeuk-Yin Chan, Eunyee Koh, and Nesreen Ahmed.
   A ML-based Approach for HTML-based Style Recommendation.
   Companion Proceedings of the ACM Web Conference, 2023 ★

4. Peter Xenopoulos, Gromit Chan, Harish Doraiswamy, Luis Gustavo Nonato, Brian Barr, and Claudio Silva.
   GALE: Globally Assessing Local Explanations.
   ICML Workshop on Topology, Algebra, and Geometry in Machine Learning, 2022 (**Spotlight**) ★

3. Yeuk-Yin Chan, Fernando Chirigati, Harish Doraiswamy, Cláudio T Silva, and Juliana Freire.
   Querying and Exploring Polygamous Relationships in Urban Spatio-Temporal Data Sets.
   *ACM International Conference on Management of Data (**SIGMOD**)*, 2017 (**Best Demo Honorable Mention**) ★

2. Rong Zheng and Yeuk-Yin Chan.
   You Like What You Hear: Using Movie Content to Improve Movie Recommender Systems.
   Winter Conference on Business Intelligence, 2016

1. Abishek Puri, Dongyu Liu, Shaoyu Chen, Siwei Fu, Tianyu Wang, Yeuk-Yin Chan, and Huamin Qu.
   ParkVis: A visual analytic system for anomaly detection in DinoFun World.
   IEEE VIS Workshop on VAST Challenge, 2015

TECH TRANSFERS

1. Audience Agent (Text-to-"Segment Definition" modeling). Adobe Experience Platform 2025, 2026.
2. AI Assistant (Text-to-"Knowledge Graph SQL" modeling). Adobe Experience Platform 2024.
3. Intelligent Captions in Mobile Scorecards. Adobe Analytics 2024.
4. HTML Email Converter. Adobe Journey Optimizer 2023.
5. Intelligent Captions for CDP Insights. Adobe Experience Platform 2022.

PATENTS

1. Data Exploration Using Natural Language With Data Sampling. US Patent App. 19/691,413.
2. Multi-Agent Design System. US Patent App. 19/662,166.
3. Axis-Based Variety In Generative Artificial Intelligence. US Patent App. 19/650,581.
4. Scalable Usability Testing System Using Simulation Agents. US Patent App. 19/459,837.
5. Generating Scalable Vector Graphic Images Bound To Insight-Backing Data. US Patent App. 18/933,259.
6. Modularized And Extensible Framework For Visualization-To-Caption Generation. US Patent App. 18/663,531.
7. Natural Language Query Filtering. US Patent App. 18/747,630.
8. Digital Content Generation From A Text-Based Input. US Patent App. 18/594,376.
9. Generation Of Data Story Recommendations Via Elicited User Feedback. US Patent App. 18/422,665.
10. Generating Hierarchical Queries From Natural Language. US Patent App. 18/486,603.
11. Performing Machine Learning Techniques For Hypertext Markup Language-Based Style Recommendations. US Patent App. 18/225,906.
12. Hypergraph Representation Learning. US Patent App. 18/358,502.
13. Generating Data Insights. US Patent App. 18/328,980.
14. Approximate Query Processing For Join Queries. US Patent App. 18/092,779.
15. Generating Editable Email Components Utilizing A Constraint-Based Knowledge Representation. US Patent App. 18/055,238.
16. Systems And Methods For Document Generation. US Patent App. 17/809,371.
17. Extensible Framework For Generating Accessible Captions For Data Visualizations. US Patent App. 17/656,254.
18. Trait Expansion Techniques In Binary Matrix Datasets. US Patent App. 17/677,323.

19. Dynamic Clustering Of Sparse Data Utilizing Hash Partitions. US Patent App. 16/852,110.
20. Segmenting Users With Sparse Data Utilizing Hash Partitions. US Patent App. 17/660,328.

TEACHING

2021 Spring	CS-GY 9223 Visualization for Machine Learning (15 students)
2020 Spring	CS-GY 9223 Visualization: Connections with Machine Learning (23 students)
2018 Fall	CS-GY 6313 Information Visualization (46 students)
2016 Fall	CS-UY 1134 Data Structures and Algorithms (51 students)

ADVISING/MENTORING

Ph.D. Committee Member

- Huayi Wang (Advisor: Jun Xu), Georgia Institute of Technology

Primary Internship Mentor:

- Chu Li (UW) - Studio Environment for Human-AI Co-creation, 2026
- Litao Yan (UPenn) - Sensemaking Through Text-to-SQL, 2026
- Shishi Xiao (Brown) - Visual Blending for Infographics Authoring, 2025
- Kevin Scroggins (UF) - Persona driven website interaction simulation, 2025
- Yuzhe You (Waterloo) (with Tongyu Zhou) - Agentic Creative Design Authoring, 2025
- Steffen Holter (ETH) (with Doga Dogan) - Operator for UX Testing, 2025
- Donghoon Shin (UW) - Generative Agent for HCI Systems, 2024
- Tongyu Zhou (Brown) - Text Based Infographics Authoring, 2023
- Zhihua Jin (HKUST) - Low Code Vis Programming, 2023
- Huayi Wang (Gatech) - ML Query Acceleration, 2022
- Pat Sukprasert (Northwestern) - Dataset Discovery, 2021

Secondary Internship Mentor:

- Kexue Fu (CityU), 2026
- Ayana Monroe (Cornell), 2024
- Ashley Zhang (UMich), 2024
- Xinyu Shi (Waterloo), 2024
- Chen Chen (UMD), 2022
- April Chen (Harvard), 2022
- Alex Tang (Northwestern), 2022
- Guande Wu (NYU), 2022

PROFESSIONAL SERVICES

Conference Organization:

- *Supporter Chair*: IEEE VIS 2027
- *Systems Track Chair*: VLDB DASHSys Workshop 2026

Program Committee:

- ACM Conference on Human Factors in Computing Systems (CHI), 2027
- Conference on Artificial Intelligence (AAAI), 2027
- IEEE Visualization Conference (VIS), 2026
- ACM Designing Interactive Systems (DIS), 2026

- Annual Meeting of the Association for Computational Linguistics (ACL) Industry Track, 2026
- ACM Web Conference (WWW), 2026
- International Conference on Web Search and Data Mining (WSDM), 2024
- ACM Web Conference (WWW), 2024
- International Conference on Web Search and Data Mining (WSDM), 2023
- ACM Web Conference (WWW), 2023
- International Conference on Web Search and Data Mining (WSDM), 2023
- ACM Web Conference (WWW), 2022
- International Conference on Web Search and Data Mining (WSDM), 2022
- Conference on Information and Knowledge Management (CIKM) Demo, 2022

Reviewer:

- ACM CHI (since 2021), IEEE VIS (since 2019), WWW (since 2020), EuroVis (since 2017), PacificVis (since 2020), IEEE CG&A, IEEE TVCG

PRESENTATION, NEWS AND MEDIA COVERAGE

2026 *Designing Visualization Computations for Human Perception and Creativity*
Guest Lecture, Boston University (Host: Prof. Chang Xiao)

2026 *Designing Visualization Computations for Human Perception and Creativity*
Seminar, UC Davis (Host: Prof. Kwan-Liu Ma)

2025 *LLM Driven Visualization Creation*
VisInPractice, IEEE VIS 2025

2024 *Visualization Authoring Meets Large Language Model*
Guest Lecture, New York University (Host: Prof. Qi Sun)

2024 *Project Infograph It (a.k.a. Epigraphics)*
Adobe Summit Sneaks (Press Coverage: Fast Company, ZDNET)

2022 *Project Style Blast*
Adobe Summit Sneaks (Press Coverage: PC Watch, GSPANN)

2021 *Interactive Audience Expansion On Large Scale Online Visitor Data*
KDD, Singapore

2021 *Data Summaries for Scalable Visual Analysis*
Dataminr, New York; Megagon Labs, Mountain View; Adobe Research, San Jose

2020 *Real-Time Clustering for Large Sparse Online Visitor Data*
The Web Conference, Taipei

2019 *Visualizing and Understanding Complex Upper Limb Movement Under Obstetrical Brachial Plexus Injuries*
IEEE VIS, Vancouver

2018 *Visualizing Bipartite Relations at Scale with the Minimum Description Length Principle*
AT&T Graduate Student Symposium, New York City; IEEE VIS, Berlin

2016 Student Hackers Team Up with Manhattan DA to Fight Human Trafficking

2016 科網雲圖：財經新聞大數據 (in Chinese)

2016 科大生研大數據獲邀海外演講 (in Chinese)

AWARDS/HONORS/CERTIFICATIONS

2024 CHI Best Paper Honorable Mention

2023 INLG Best Paper

2018 VIS Student Volunteer

2017 SIGMOD Best Demo Honorable Mention

2017 SIGMOD Student Travel Grant
2016 NYU School of Engineering Fellowship
2016 Bronze Prize of ASM Technology Award
2016 Mr Armin and Mrs Lillian Kitchell Undergraduate Research Award
2016 Technology and Management Elite Student Scholarship
2015 Honorable Mention, IEEE VAST (Visual Analytics Science and Technology) Challenge 2015
2013 HKSAR Government Scholarship Fund - Reaching Out Award
2012 University Entrance Scholarship, HKUST
2012 Morrison Distinction Price, Queen's College